

control capabilities in an affordable, compact package. MicroLogix 1200 provides features and options to handle an extensive range of applications. The expandable MicroLogix 1500 controller helps you achieve high-level control in a variety of applications. MicroLogix 1100 and MicroLogix 1400 controllers increase application coverage with enhanced network communications at affordable prices. RSLogix 500 programming



Fig. 2: MicroLogix 1000 PLC (Model 1761-L10BWA) features 120V/240V AC power, six 24V DC digital inputs and four relay outputs)



Fig. 3: MicroLogix 1400 PLC (Model 1766-L32BWA) features 12 digital fast 24V DC inputs, eight digital normal 24V DC inputs, 12 relay outputs, RS232 ports, 110V/240V AC power)

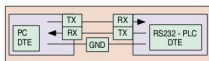


Fig. 4: NULL modem connections

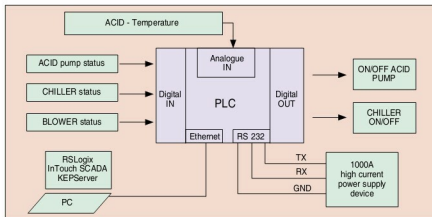


Fig. 5: Schematic diagram for PLC connections

software and RSLogix Micro programming software provide an instruction set that is common to MicroLogix and SLC 500 controller families.

#### MicroLogix 1000 PLC system.

MicroLogix 1000 PLC (Fig. 2) is available in 10-point, 16-point or 32-point digital input/output (I/O) versions. Analogue versions available with 20 digital I/O points, four analogue inputs (two voltage and two current) and one analogue output (configurable for either voltage or current) provides a compact form factor, with footprints as small as 120mm x 80mm x 40mm (4.72 x 3.15 x 1.57 inches).

It offers fast processing with typical throughput time of 1.5ms for a 500-instructions program and has a pre-configured 1kB program and data memory to ease configuration. It includes built-in EEPROM memory; hence, there is no need for battery back-up or a separate memory module.

The device offers peer-to-peer messaging (up to 32 controllers on a DH-485 network) through a 1761-NET-AIC communication interface. Communication takes place via DeviceNet and EtherNet/IP through 1761-NET-DNI and 1761-NET-ENI communication interfaces.

It also includes a built-in high-speed counter (only on controllers with 24V DC inputs) and lets you customise input response times and noise rejection using adjustable DC input filters.

The device supports simple connectivity through an RS232 communication channel to a PC for program upload, download and monitoring.

#### MicroLogix 1400 PLC system.

The 1766 MicroLogix 1400 PLC system (Fig. 3) is built upon critical MicroLogix 1100 features, including EtherNet/IP, online editing and a built-in LCD panel. These controllers feature a higher I/O count, faster high-speed counters, pulse train output, enhanced network capabilities and a backlight on the LCD panel. Controllers without embedded analogue I/O points provide 32 digital I/O points, while analogue versions offer 32 digital I/O points and six analogue I/O points. You can expand all versions with up to seven 1762 expansion I/O modules.

The Ethernet port provides Web server capability, email capability and protocol support for DNP3 protocol support. The built-in LCD with a backlight lets you view the controller and I/O status. It also provides a simple interface for messages, bit/integer monitoring and manipulation.

Application capabilities can be expanded through support for up to seven 1762 MicroLogix Expansion I/O modules with 256 discrete I/Os, up to six embedded 100kHz high-speed counters (only on controllers with DC inputs), two serial ports with DF1, DH-485, Modbus RTU, DNP3 and ASCII protocol support.

There are 10kB words in the user program memory with 10kB words in the user data memory, and up to 128kB for data logging and 64kB for recipe.

### PLC connections

One can buy a compact AB Allen Bradley MicroLogix 1400 PLC 1766, with eight DIO and two built-in serial ports (model 1766) for initial development. We used MicroLogix 1400 controllers, which are suitable for use in an industrial environment. Specifically, this equipment is intended for use in clean, dry environments.

Typical PLC connections for automation of an experimental facility are shown in Fig. 5. Allen Bradley Micro